

SDLC Models, Agile Scrum & Requirements Engineering - Detailed

DETAILED COMPREHENSIVE NOTES

Subject: Software Engineering | Chapter: Chapter 1: Software Development Life Cycles & Agile Methodologies | Academic Year: 2026-2027

1. Software Development Life Cycle (SDLC) Framework

The Software Development Life Cycle (SDLC) defines a systematic process for building software: Requirements Analysis, System Design, Implementation, Testing, Deployment, and Maintenance.

Waterfall Model: Linear-sequential model suitable when requirements are clear and immutable. Disadvantage: late defect discovery during testing.

Spiral Model: Risk-driven iterative model structured in four quadrants (Determine Objectives, Identify & Resolve Risks, Develop & Test, Plan Next Phase).

2. Agile Principles & The Scrum Framework

The Agile Manifesto prioritizes: Individuals and interactions over processes and tools; Working software over comprehensive documentation; Customer collaboration over contract negotiation; and Responding to change over following a plan.

Scrum is an empirical Agile framework delivering software in fixed-length iterations called Sprints (typically 1-4 weeks).

Scrum Roles: Product Owner (manages backlog priorities), Scrum Master (removes blockers, facilitates process), and Development Team (cross-functional creators).

- * Sprint Planning: Selects Product Backlog Items (PBIs) and defines the Sprint Goal.
- * Daily Standup: 15-minute sync: What did I do yesterday? What will I do today? What blockers exist?
- * Sprint Review: Demonstrates working software to stakeholders.
- * Sprint Retrospective: Team inspects internal processes to identify continuous improvements.

3. Requirements Engineering & User Story Modeling

Requirements engineering bridges stakeholder needs and technical specifications.

Functional Requirements define what the system must do; Non-Functional Requirements define quality attributes (performance, security, scalability, availability).

User Story format: "As a [type of user], I want [an action/goal] so that [a benefit/value]."

INVEST criteria for effective user stories: Independent, Negotiable, Valuable, Estimable, Small, and Testable.